Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

WEEKLY LESSON PLAN

Department: Computer Science and Engineering

Year / Semester: 2025-26 / Winter

Course Code / Course Name: 10212CS292 / Applied Programming Skills

Unit V: Algorithm Design Techniques
Course Outcome: Develop algorithmic solutions using programming strategies for optimization problems.

Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 9	Session 41	Divide and Conquer pattern	Task 71: **Majority Element** Task 72: **Reverse Bits** Task 73: **Longest Nice Substring**	Leet Code
	Session 42	Merge sort for counting inversions	Task 74: **Count of Smaller Numbers After Self**	
	Session 43	Quick select (Kth element)	Task 75: **K Closest Points to Origin**	
	Session 44	Dynamic Programming	Task 76: **Climbing Stairs**	
	Session 45	One Dimensional (house robber, climbing stairs)	Task 77: **House Robber** Task 78: **Min Cost Climbing Stairs**	

Week	Session	Topics to be Covered	Problems to Practice (Hands-On)	Platform
Week 10	Session 46	Two Dimensional (grid paths)	Task 79: **Unique Paths**	Leet Code
	Session 47	Subset Dynamic Programming (0/1 knapsack style)	Task 80: **Ones and Zeroes** Task 81: **Coin Change**	
	Session 48	Greedy interval scheduling	Task 82: **Minimum Number of Arrows to Burst Balloons**	
	Session 49	Backtracking (subsets, permutations, combinations)	Task 83: **Subsets** Task 84: **Permutations** Task 85: **Combinations**	
	Session 50			